Aquacultural Engineering 84 (2019) 12–22

Contents lists available at ScienceDirect

Aquacultural Engineering

journal homepage: www.elsevier.com/locate/aque

A bioenergetic and protein flux model to simulate fish growth in commercial farms: Application to the gilthead seabream

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ARTICLE INFO

Keywords:
Bioenergetic model
Dynamic protein mass balance
Aquaculture production model
Fish farming
Gilthead seabream
Sparus aurata

ABSTRACT

A model for fish growth simulation based on the bioenergetic factorial approach is presented. This work presents a novel approach that extends the traditional bioenergetic model by explicitly including the Energy and Protein fluxes (EP model). This is a valuable feature that allows the dynamic simulation of fish proximate composition. For the aquaculture industry it represents a trade-off between detailed process simulation and feasibility of model implementation, namely regarding data gathering on an operational setting. The EP model is targeted to simulate fish production in commercial farms. Farm data for feed intake, feed composition (energy and protein content), temperature over time and the initial fish body weight are the only required data to run the model. Furthermore, apparent digestibility coefficient (ADC) values of the feed used in the farm must be known or else ADC values of feeds with similar composition can be used. The EP model implementation is illustrated for the gilthead seabream (*Sparus aurata*) based on published experimental data. The model was validated ($r = 0.997$, $p < 0.05$, $n = 12$) using a published experimental data set for gilthead seabream reared in a range of temperatures that reproduce the conditions in most countries producing this species. For the entire growth period (488 days) the estimated mean absolute error (MAE) is 8.8 g.fish^{-1} and the mean absolute percentage error (MAPE) is 8.3%. Simulation of fish growth in real operational conditions is evaluated with three datasets from a commercial farm that operates in earthen ponds with temperatures ranging between 12.6 °C and 24.8 °C. Overall the model outputs match well with the production data in the 3 batches. Initial weight ranged between 2.8 g and 3.7 g. The deviation between the data and simulated final weights is below 15 g, for a final weight around 435 g. The maximum absolute error is 21.1 g per fish (*MAE*) and in percentage 8.3% (*MAPE*).

1. Introduction

Aquaculture farmers need to control input resources, efficiency, variables affecting it, disease outbreaks and waste production whilst assuring real-time management of the process. Simulation models can be a supporting tool for understanding, predicting and managing

Abbreviations: Δt, timestep; x, model variables generalized for energy (E) and protein (P); ADC, apparent digestibility coefficient; BW, individual body weight; BWt, individual body weight at any given point in time; *COG_x*, costs of growth; DEB, dynamic energy budget; *DI_x*, digestible intake; E, energy; EP model, The Energy and Protein fluxes model; FCR, feed conversion ratio; $Fish\_P_t$, Fish protein content at any given point in time; $Fish\_L_t$, Fish lipid content at any given point in time; $Fish\_moist_t$, Fish moisture content at any given point in time; $Fish\_ash_t$, Fish ash content at any given point in time; *FM_x*, fasting metabolism requirements; *FM_coef_x (T)*, fasting metabolism coefficient as a function of water temperature; *FM_coef_x*, fasting metabolism coefficient at a given temperature; *G_P*, protein gain rate; *G_L*, lipid gain rate; *G_x*, gain; *L_EnergyContent*, the energy value of lipids; MAE, mean absolute error; MAPE, mean absolute percentage error; MBW, metabolic body weight; *mbw_exp_x*, exponent of the metabolic body weight; p, probability of the null hypothesis; P, protein; *P_EnergyContent*, the energy value of protein; r, Pearson's coefficient; $specific\_MoistContent(BW_{t-\Delta t})$, specific fish moisture content dependent on fish weight; *specific_AshContent*, specific fish ash content; t, time; T, water temperature

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https://doi.org/10.1016/j.aquaeng.2018.11.001
Received 9 February 2018; Received in revised form 30 October 2018; Accepted 9 November 2018
Available online 14 November 2018

aquaculture systems (Cuenco, 1989; Pomeroy et al., 2008). State of the art modelling approaches that simulate fish growth include nutrient-based models (Dumas et al., 2010) and the dynamic energy budget (DEB) (Nisbet et al., 2012; Serpa et al., 2013; Stavrakidis-Zachou et al., 2018). However, a model suitability depends on the purpose of its application. If the simulation is at the farm level and the objective is to simulate current conditions to provide real-time outputs to the operators, it may be unfeasible to obtain detailed data able to feed such models, or the advantages of detailed models, might dissipate with the complexity for its setup.

A simpler approach to simulate fish growth is the traditional bioenergetic model (e.g. Brigolin et al., 2010). There are many published models using the traditional bioenergetic approach that apply different formulations. Some include a generic approach, such as the Wisconsin modelling initiative, which focuses on simulation of wild stocks (Hanson et al., 1997). Canale and Breck (2013) analysed a wide range application of bioenergetics models and concluded that its incorrect formulation could lead to large mathematical errors. When formulating bioenergetic models conservation of energy must be ensured. Besides the model formulation its parameterization for a given species is also important for the consistent and general model application. The approach developed by Lupatsch (2003) and Lupatsch et al. (2001, 2003a, 2003b) not only includes consistent growth and metabolic experiments, but also generates the required data to parameterize a generic model for a given species. Lupatsch (2003) designed a set of experiments that allow the consistent quantification of the following energy and protein fluxes as a function of body weight: feed intake, digestible feed intake (discounts faecal losses), fasting metabolism and retention. In a related work by Oliva-Teles et al. (2011) the authors present equations for maintenance requirements and retention as a function of temperature (besides body weight).

The goal of this work is to formalize a generic growth model based on the factorial approach undertaken by Lupatsch (2003) and Lupatsch et al. (2001, 2003a, 2003b) that requires minimal dataset for application in an operational setting. This approach is herein named Energy and Protein flux (EP) model. The EP model extends the traditional bioenergetic approach by explicitly coupling the energy and protein fluxes and translating those into individual weight gain and proximate composition simulation over time. Further objectives of this paper are to i) exemplify this model parameterization for the gilthead seabream (*Sparus aurata*) using the work by Lupatsch (2003), Lupatsch et al. (2003a) and Oliva-Teles et al. (2011), ii) validate the model with an independent dataset, and iii) evaluate its practical application with a commercial farm dataset.

2. Methodology

2.1. EP model description

The EP model dynamically simulates the fluxes of energy and protein, the model outputs are proximate composition and individual body weight (BW_t) at any given point in time (t) considering the initial value for BW and the data inputs for temperature, feed intake, energy/protein feed content and digestibility.

The BW_t is a model output and is given as the sum of protein, lipids, water and ash contents (Eq. 1):

$$BW_t = Fish\_P_t + Fish\_L_t + Fish\_moist_t + Fish\_ash_t \quad (1)$$

Where:

i) Fish protein and lipid content ($Fish\_P_t$, in g P.fish^{-1} and $Fish\_L_t$, in g L.fish^{-1}, respectively) are the model state variables and are given by the differential equations as in Eqs. (2) and (3), based on the respective gain rates (G_P in g P.fish^{-1}.d^{-1}; G_L in g L.fish^{-1}.d^{-1}). The fish lipid gain is calculated as in Eq. (4), based on the assumption that fish energy gain is the sum of the energy provided by protein and lipids, and that the energy value of protein and lipids (*P_EnergyContent* in kJ.g P^{-1} and *L_EnergyContent* in kJ.g L^{-1}, respectively) are constants.

$$dFish\_P/dt = G\_P \quad (2)$$

$$dFish\_L/dt = G\_L \quad (3)$$

$$G\_L = (G\_E - G\_P * P\_EnergyContent)/L\_EnergyContent \quad (4)$$

The G_E and G_P are described based on the energy and protein balance, respectively, as described in detail in *2.1.1 Bioenergetic and protein balance* section.

ii) Fish moisture ($Fish\_moist_t$, in g.fish^{-1}) and fish ash content ($Fish\_ash_t$, in g.fish^{-1}) need to be parameterized per species as a function of fish weight, which can be determined experimentally (Lupatsch, 2003). As an approximation, calculation of fish moisture and ash content in time t ($Fish\_moist_t$ and $Fish\_ash_t$) is based in fish BW in a previous timestep (Δt), as in Eqs. (5) and (6).

$$Fish\_moist_t = Fish\_moist_t'(BW_{t-\Delta t}) \quad (5)$$

$$Fish\_ash_t = Fish\_ash_t'(BW_{t-\Delta t}) \quad (6)$$

The glycogen content of the fish, which can range from 1% to 2% per dry matter of fish tissue (about 0.5% per wet weight of fish), is not included in this model. This is a commonly adopted assumption in fish modelling (Canale and Breck, 2013).

2.1.1. Bioenergetic and protein balance

Herein, is used a factorial approach of feed and energy partitioning in fish (Fig. 1) that includes the components that can be experimentally parameterized for a given species and operationally simulated at a farm.

The mass balance is generalized for both energy (E) and protein (P) using *_x*, as formulated in Eq. (7):

$$DI\_x = FM\_x + COG\_x + G\_x \quad (7)$$

Where, DI_x is digestible intake, FM_x is the fasting metabolism requirements, COG_x represents the costs of growth and G_x is gain. All these fluxes are expressed as kJ.fish^{-1}.d^{-1} and g P.fish^{-1}.d^{-1}, for E and P, respectively. In this work the mass balance is re-written to estimate retention for E and P (Eq. (8)):

$$G\_x = DI\_x - FM\_x - COG\_x \quad (8)$$

DI_x represents total feed intake minus faecal losses. DI_x is a model input that can be calculated from data about: i) total feed distributed to fish at the farm (assuming that all is consumed or that a certain % is lost), ii) number of fish, iii) feed energy and crude protein contents, and iv) apparent digestibility coefficients (ADC) for energy and crude protein, which are feed and species specific (Lupatsch et al., 1997).

There are several metabolic rates (basal, standard, routine, active) and distinct methods for measuring each one (Strand, 2009; Jobling, 2011). This model includes the energy and protein losses at fasting as a proxy for the fasting metabolic rate (FM_x). Generally, the metabolic rate per fish is defined as a function of BW and temperature (Hanson et al., 1997; Lupatsch, 2003; Oliva-Teles et al., 2011), Eq. (9):

$$FM\_x = FM\_coef\_x(T) * BW^{mbw\_exp\_X} \quad (9)$$

$FM\_coef\_x(T)$ is the fasting metabolism coefficient as a function of water temperature (T, in °C) and is expressed in kJ.kg$^{-mbw_exp_E}$.d^{-1} or g P.kg$^{-mbw_exp_P}$.d^{-1}, for E and P, respectively. FM_coef_x can be experimentally estimated for a given temperature by the loss of energy and protein at fasting, i.e. when DI_x equals to zero, as illustrated in Fig. 2 (Lupatsch, 2003; Oliva-Teles et al., 2011). Based on experiments for several temperatures, $FM\_coef\_x(T)$ might be defined based on data fitting. $BW^{mbw\_exp\_X}$ is the metabolic body weight (MBW), where mbw_exp_x is the exponent of the metabolic body weight for energy and protein and can be calculated from plots of energy and protein loss

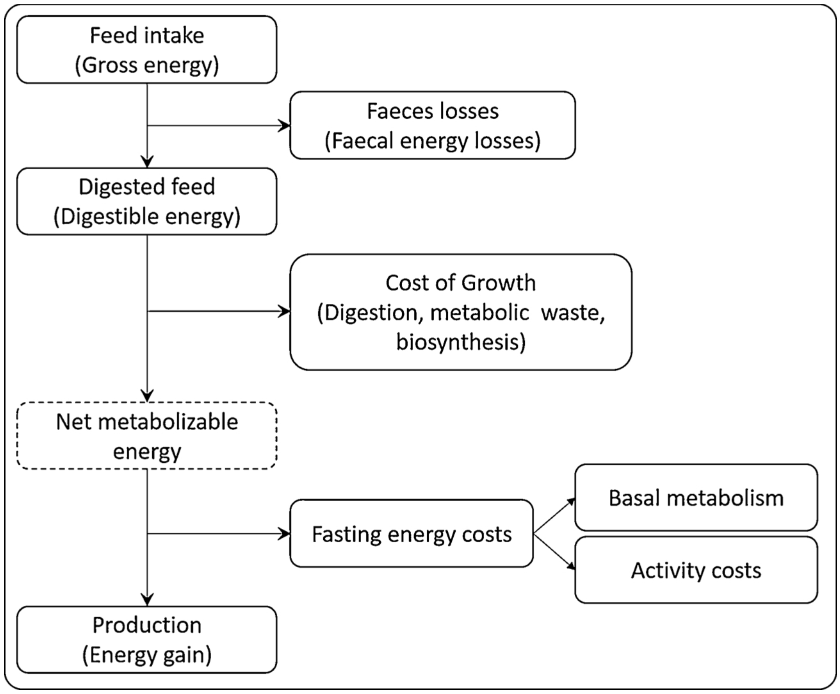

Fig. 1. Factorial diagram of feed and energy partitioning in fish (adapted from Brett, 1995; Cho and Kaushik, 1990; Conceição et al., 1998a; FAO, 2003; Jobling, 2011; NRC, 2011).

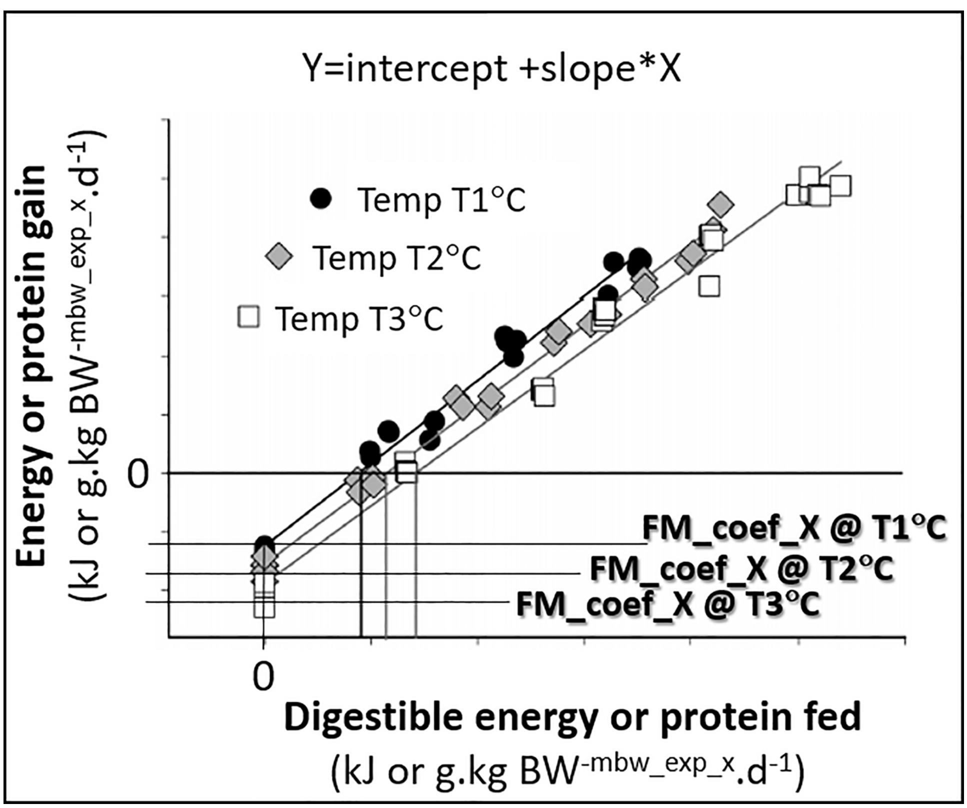

Fig. 2. Calibration procedure: Plots of energy or protein gain (expressed as kJ or g. kg$^{-mbw_exp_x}$.d^{-1}) at different rearing temperatures *vs* digestible intake also expressed per metabolic body weight (kg $^{-mbw\_exp\_x}$) adapted from Oliva-Teles et al. (2011).

($kJ.fish^{-1}.d^{-1}$ or $g.fish^{-1}.d^{-1}$) vs *BW* ($kg.fish^{-1}$) as illustrated by Lupatsch (2003).

COG_x represents the costs of processing and depositing nutrients into body tissues, and in particular the costs of protein synthesis/deposition (Houlihan et al., 1995; Conceição et al., 1998a; Bureau et al., 2000). This model term includes a multiplicity of physiological processes namely digestion or sometimes referred to as specific dynamic action, metabolic wastes and biosynthesis. *COG_x* cannot be directly quantified. It must be derived based on the experimentally measurable components of the mass balance (i.e., *DI_x, FM_x* and *G_x*), which ensures that the mass balance is closed (Fig. 1). *COG_x* parameterization is illustrated in this paper (*2.2.3 Costs of growth, COG_x*) for the gilthead seabream.

This model can be further extended to encompass other activity energy costs beyond the standard, e.g. due to increased activity caused by an increase of current velocity in cages or due to increased swimming activity caused by a stress event.

To simulate the fish protein and lipid content ($Fish\_P_t$ and $Fish\_L_t$) and *BW* over a production cycle for a given species it is required to parameterize:

i) The *FM_x* and *COG_x* functions of both the bioenergetic and protein balances;
ii) The $Fish\_moist_t$ and $Fish\_ash_t$ functions, i.e., fish moisture and ash functions of fish weight.

In the next section is illustrated this procedure for the gilthead seabream.

2.2. Model implementation for the gilthead seabream

In this paper the authors illustrate the model implementation for the gilthead seabream. The model was parameterized with data and coefficients from Lupatsch (2003), Lupatsch et al. (2003a) and Oliva-Teles et al. (2011) as systematised in Table 1 and detailed here.

The model differential equations, Eqs. (2) and (3), are solved with Euler method with a timestep (Δt) of 1 day. To run the model for each data set it was considered the initial *BW* and as data inputs the interpolated water temperatures and registered feed intake over time. The feed intake was converted into digestible energy intake (*DI_E*) and into digestible crude protein intake *DI_P*, based on the feed nutritional information (energy and crude protein contents) and ADC values. Model initial conditions and data inputs for running each data set are presented in Table 2.

The model calibration was carried out by comparing raw data published by Lupatsch (2003, Annex 1 and 2) and the simulated results, for *BW* and fish energy, protein and lipid content. The outputs of the gilthead seabream model applied to the dataset used for model parameterization are shown in the results section.

The validation was carried out using data from an experimental dataset presented by Zurita (2013). The model was further applied to simulate gilthead seabream *BW* over time reared in 3 ponds of a commercial farm. Each pond has an initial stock of about 310 to 350 million fish. The farm operates in earthen ponds and is located in southwest Europe (in Ria de Alvor, Portugal). The outputs of the model validation and application are shown in the results section.

The comparison of the model results with the data was evaluated by calculating the Pearson's coefficient (r) to estimate the level of correlation between the data and model outputs for individual biomass and the t-test to estimate its significance (probability of the null hypothesis, p). For the validation datasets the mean absolute error (MAE) and the mean absolute percentage error (MAPE) were also calculated regarding individual biomass.

Furthermore, the model was audited to ensure the proper use of the mass balance as detailed by Canale and Breck (2013). As recommended by these authors, the change in the fish energy content was verified against the total net energy provided (Eq. (3) in Canale and Breck, 2013), which resulted in consistent outputs.

The parameterization of the $Fish\_moist_t$, $Fish\_ash_t$, *FM_x* and *COG_x* functions is described next.

2.2.1. Fish moisture and ash contents, $Fish\_moist_t$ and $Fish\_ash_t$

Following Lupatsch (2003) $Fish\_moist_t$ ($g.fish^{-1}$) for gilthead seabream is calculated considering a specific fish moisture content dependent on fish weight ($specific\_MoistContent(BW_{t\text{-}\Delta t})$, in $g.\ kg\ fish^{-1}$), as in Eq. (10); and $Fish\_ash_t$ ($g.fish^{-1}$) is calculated considering a constant specific fish ash content (*specific_AshContent*, in $g.kg\ fish^{-1}$), as in Eq. (11):

$$Fish\_moist_t = specific\_MoistContent(BW_{t\text{-}\Delta t}) * 10^{-3} * BW_{t\text{-}\Delta t} \qquad (10)$$

$$Fish\_ash_t = specific\_AshContent * 10^{-3} * BW_{t\text{-}\Delta t} \qquad (11)$$

Expression and value for $specific\_MoistContent(BW_{t\text{-}\Delta t})$ and *specific_AshContent* are provided in Table 1.

2.2.2. Fasting metabolism requirements, FM_x

The *mbw_exp_x* has previously been estimated by Lupatsch (2003) and Lupatsch et al. (2003a). According to their findings the *mbw_exp_x* is different for the energy and protein metabolism which *"reflects the difference in the relationships between energy and protein and increasing body weight, where percent protein is generally conserved in regard to fish size."* (Lupatsch, 2003). The values adopted in the EP gilthead seabream model follow the ones adopted by Lupatsch et al. (2003a) and Oliva-Teles et al. (2011) and correspond to 0.80 and 0.70 for energy and protein, respectively (Table 1). The *FM_coef_x* was calculated based on the linear regressions between digestible energy fed, in $kJ.Kg\ fish^{-0.80}.d^{-1}$ (and protein fed in $g.Kg\ fish^{-0.70}.d^{-1}$) and the energy gain, in $kJ.Kg\ fish^{-0.80}.d^{-1}$ (and protein gain in $g.Kg\ fish^{-0.70}.d^{-1}$) presented

Table 1
EP Model parameterization for the gilthead seabream.

Parameter/function	Units	Value	Source
P_EnergyContent	($kJ.g\ P^{-1}$)	23.1 ± 0.84	Lupatsch et al. (2003a).
L_EnergyContent	($kJ.g\ L^{-1}$)	35.8 ± 1.29	Lupatsch et al. (2003a).
$specific\_MoistContent(BW_{t\text{-}\Delta t})$	($g.\ kg\ fish^{-1}$)	$584(\pm 6.0)*(BW*10^{-3})^{-0.043(\pm 0.003)}$	Eq. (8) in Lupatsch (2003).
specific_AshContent	($g.\ kg\ fish^{-1}$)	43.3(± 5.3)	Lupatsch (2003).
FM_coef_E(T)	($kJ.Kg\ fish^{-mbw\_exp\_E}.d^{-1}$)	$7.43*e^{T*0.068}$	Data fitting (Oliva-Teles et al., 2011) detailed in Table 2.
FM_coef_P(T)	($g.Kg\ fish^{-mbw\_exp\_P}.d^{-1}$)	$0.061*e^{T*0.068}$	Data fitting (Oliva-Teles et al., 2011) detailed in Table 2.
mbw_exp_E	(-)	0.80	Lupatsch et al. (2003a); Oliva-Teles et al. (2011).
mbw_exp_P	(-)	0.70	Lupatsch (2003); Lupatsch et al. (2003a); Oliva-Teles et al. (2011).
COG_coef_E	(-)	0.33	Oliva-Teles et al. (2011).
COG_coef_P(BW)	(-)	IF $BW < 30\ g \rightarrow 0.5$ IF $BW < 50\ g \rightarrow 0.55$ IF $BW >= 50\ g \rightarrow 0.6$	Calibration in this study, based on Lupatsch (2003) data and Oliva-Teles et al. (2011) equations.

Table 2
Simulation initial conditions and data inputs.

Data source	Datasets	Initial value BW*2	Data inputs: Temperature	Feed intake	Feed composition	Feed digestibility
Lupatsch, 2003 (Parameterization)	13 trials	0.72 g to 197.0 g	Interpolation of weekly records	Weekly records	460 g protein.kg feed^{-1}; 18.9 MJ kg feed^{-1}	78% for energy, 83% for crude protein
Zurita, 2013 (Validation)	1 production cycle	10.01 g	Interpolation of interspersed records (488 day cycle with total of 15 records)		450 g protein.kg feed^{-1}; 18.4 MJ kg feed^{-1}	78% for energy, 83% for crude protein *1
Commercial farm data (Validation)	3 production cycles	2.8 g to 3.7 g	Daily records		From feed bags	*1

*1 Assumed ADC values from Lupatsch and Kissil (1998) and Lupatsch (2003), for commercial feed with similar composition.
*2 Initial value of variables from Eq. (1) are calculated based on initial BW (IBW) multiplied by specific fish content for protein, lipids, water and ash contents, as defined by Lupatsch (2003):
initial $Fish\_P_t \longleftarrow 177\ (g.\ kg\ fish^{-1}) * 10^{-3} * IBW$;
initial $Fish\_L_t \longleftarrow 217 * (IBW * 10^{-3})^{0.225} * 10^{-3} * IBW$;
initial $Fish\_moist_t \longleftarrow specific\_MoistContent(BW_{t\text{-}\Delta t}) * 10^{-3} * IBW$;
initial $Fish\_ash_t \longleftarrow specific\_AshContent * 10^{-3} * IBW$.
$specific\_MoistContent(BW_{t\text{-}\Delta t})$ and *specific_AshContent* are provided in Table 1.

Table 3
FM_coef_x per temperature and temperature dependence function FM_coef_x(T).

Temperature (°C)	*FM_coef_E* (kJ.Kg fish$^{-mbw_exp_E}$.d^{-1})	*FM_coef_P* (g.Kg fish$^{-mbw_exp_P}$.d^{-1})	Source
2.0	Lethal limit		Sola et al. (2007).
16.0	25% less relative to 21 °C ≈ 0.75*30.2	–	Guinea and Fernandez (1997) to evaluate the exponential function.
20.5	30.2	0.25	Oliva-Teles et al. (2011).
24.0	38.1	0.30	
27.0	47.1	0.39	
33.0	Lethal limit		Sola et al. (2007).
FM_coef_x(T):	$7.43*e^{T*0.068}$	$0.061*e^{T*0.068}$	Data fitting (Oliva-Teles et al., 2011).

Table 4
Calibration of *COG_coef_x* based on the linear regressions presented in equations from Oliva-Teles et al. (2011; eqs. 7.6 to 7.8 and eqs 7.9 to 7.11) and from Lupatsch (2003; eqs. 19, 22 and 23).

		COG_coef_E	*COG_coef_P*
Oliva-Teles et al. (2011)	20.5 °C	0.31	0.54
	24.0 °C	0.33	0.54
	27.0 °C	0.34	0.52
	Average	0.33	0.53
Lupatsch (2003)	Below and at maintenance		0.49
	Above maintenance		0.72
	No differentiation		0.66
Calibration of *COG_coef_x* adopted in this study:		0.33	IF $BW < 30\,g \rightarrow 0.50$ IF $BW < 50\,g \rightarrow 0.55$ IF $BW \geq 50\,g \rightarrow 0.60$

by Oliva-Teles et al. (2011) at 20.5 °C, 24 °C and 27 °C. The intercept with y-axis of the regression corresponds to the energy (and protein) loss at fasting and thus is a proxy of the *FM_coef_x* (Fig. 2).

FM_coef_x(T), was derived by fitting the exponential function to the obtained values of *FM_coef_x* at each temperature and is presented in Table 3. The fit of the *FM_coef_E(T)* exponential function was evaluated considering an additional point based on Guinea and Fernandez (1997) to verify the function applicability to an extended temperature span. From Guinea and Fernandez (1997) it is possible to roughly estimate that respiration at rest is 25% less at 16 °C when compared to 21 °C, which was here used as a proxy for energy requirements at fasting. The values used to estimate and evaluate the *FM_coef_E(T) and FM_coef_P(T)* functions are presented in Table 3. To ensure model robustness, *FM_coef_x(T)* is limited to accommodate lethal temperatures considered in Table 3.

2.2.3. Costs of growth, COG_x

Herein is illustrated the definition of the *COG_x* function based on the experimental work presented by Oliva-Teles et al. (2011) for the gilthead seabream: the energy and protein balance is rearranged and solved for *COG_x*, as in Eq. (12):

$$COG\_x = DI\_x - FM\_x - G\_x \tag{12}$$

COG_x function is simplified as a function of digestible intake, as in Eq. (14), by replacing in Eq. (12) the *G_x* with the gain functions obtained experimentally by Oliva-Teles et al. (2011) and generally illustrated in Fig. 2 ($Y = intercept + slope*X$). For unit consistency both terms of the experimentally obtained gain function ($Y = intercept + slope*X$, Fig. 2) need to be multiplied by the metabolic body weight ($BW^{-mbw\_exp\_x}$), which results in Eq. (13):

$$Y*BW^{-mbw\_exp\_x} = intercept*BW^{-mbw\_exp\_x} + slope*X*BW^{-mbw\_exp\_x} \Leftrightarrow G\_x = -FM\_x + slope*DI\_x \tag{13}$$

Finally, *G_x* in Eq. (12)is replaced by Eq. (13)and re-written as in Eq. (14):

$$COG\_x = COG\_coef\_x * DI\_x \tag{14}$$

Where *COG_coef_x* (dimensionless) is equal to 1 minus the slope of the linear regression (Fig. 2). The slope gives what Lupatsch (2003), Lupatsch et al. (2003a) and Oliva-Teles et al. (2011) define as the efficiency of digestible intake (energy or protein) for growth. Table 4 shows the data for the parameterization of *COG_coef_x* for energy and protein for the gilthead seabream.

Oliva-Teles et al. (2011) did not find any significant difference between the gain slopes for the different temperatures and thus the *COG_coef_x* is not considered as a function of temperature. Biologically, the costs of growth include the costs of nutrient processing and deposition (sometimes referred to as specific dynamic action) and include excretory losses above fasting metabolism. While these processes are normally not simulated as a function of temperature in traditional bioenergetics (e.g., Hanson et al., 1997), they can be influenced by temperature (Guinea and Fernandez, 1997; Bar and Radde, 2009). Thus, the existence of a temperature effect on the costs of growth coefficient should be further investigated and a wider range of temperatures should be tested in growth experiments. For instance, a

Table 5
Simulation of the dataset used for model parameterization (Lupatsch, 2003; Annex 1): trial characterization and evaluation of model outputs using the Pearson's coefficient (*r*) to estimate the level of correlation and the *t*-test to estimate its significance (probability of the null hypothesis, *p*) for the *n* pairs of data-simulated *BW* obtained in each trial.

Trial data:					Model evaluation:				
Trial no.	Duration (days)	Temperature range (°C)	Feed intake (g. fish^{-1}. d^{-1})	Initial BW (g)	Final BW (g): data	(simulated)	*r*	*p* (probabil. of null hypothesis)	*n*
1	127	20.7 - 23.9	0.05 to 0.56	0.72	24.47	(30.56)	0.999	0.0349	8
2	81	23.0 - 25.0	0.49 to 1.39	19.14	78.56	(88.03)	1.000	0.0049	5
3	93	21.9 - 24.1	0.71 to 1.13	27.33	97.87	(83.34)	0.999	0.0189	6
4	92	23.9 - 26.1	0.30 to 0.98	2.78	51.87	(57.23)	1.000	0.0269	5
5	91	22.0 - 24.1	0.73 to 1.59	33.35	116.00	(104.62)	1.000	0.0009	6
6	118	20.9 - 23.2	0.36 to 1.22	17.52	76.95	(71.86)	1.000	0.0001	8
7	131	20.7 - 22.7	0.06 to 0.45	1.02	24.79	(25.13)	1.000	0.0002	9
8	96	24.0 - 26.0	0.18 to 0.79	2.20	31.35	(34.81)	0.999	0.0020	6
9	109	20.2 - 22.0	0.14 to 0.90	2.34	36.65	(38.76)	1.000	0.0111	8
10	182	21.8 - 26.0	1.51 to 3.60	76.29	342.56	(339.11)	0.998	0.0046	10
11	132	22.9 - 26.0	2.18 to 4.70	197.00	424.70	(433.48)	0.999	0.0048	5
12	267	20.0 - 26.0	1.29 to 3.98	64.43	477.70	(453.86)	0.999	0.0001	11
13	100	23.0 - 24.8	2.93 to 4.48	193.57	418.00	(418.33)	0.998	0.0295	5

COG_coef_E variation between 0.31 to 0.34 (Table 4) results in a diversion in the calculation of the *COG_E* that corresponds to about 10% (at 20.5 °C compared with 27 °C). Based on the average value adopted by Oliva-Teles et al. (2011) for the energy gain slope (0.67), herein the value for *COG_coef_E* is 0.33 (Table 4). During model calibration based on data and equations from Lupatsch (2003) and Oliva-Teles et al. (2011) the costs of growth coefficient for protein (*COG_coef_P*) was defined for 3 size ranges (Table 4).

3. Results

The results of the EP gilthead seabream model include simulations of i) the Lupatsch (2003) dataset used for model parameterization, ii) the Zurita (2013) dataset used for validation, and iii) a real production setting using data from a commercial farm.

3.1. Model application to the dataset used for model parameterization

Herein are presented the outputs of the application of the gilthead seabream model to simulate the data from the 13 trials published by Lupatsch (2003, Annex 1 and Annex 2), which includes data used for model parameterization. Table 5 shows the individual outputs for each trial and Fig. 3 shows two charts with the model simulation over two of the trials: trial 1 with smaller individuals and trial 12 with a longer duration reaching larger-sized fish.

Overall, considering the data pairs of the 13 trials there is a significant ($p < 0.05$; $n = 92$) strong agreement ($r = 0.998$) between the model results and the fish *BW* data registered along the experiments (Fig. 4-a). Furthermore, the model outputs were compared with data for weight gain (expressed in g.fish^{-1}.d^{-1}) as depicted in Fig. 4-b).

Fig. 5 depicts the simulation of the fish protein and lipid content as a function of *BW* measured and the data from the extensive samples collected by Lupatsch (2003, Annex 2). As illustrated in Fig. 5 the model follows the overall trends of the lipid and energy content as a function of body weight and the simulated protein content is near the data range (model: 142–203 g. kg fish^{-1}; data: 147–199 g. kg fish^{-1}). However, the model would benefit from an improvement regarding fish nutrient content simulation for the larger sizes (> 300 g) whereby the visual comparison of the model results with the data indicate that the model underestimates the energy and lipid contents, and overestimates the protein content. On the contrary, for smaller sizes between 20 g and 90 g the model overestimates fish lipid and energy contents.

3.2. Model validation

For model validation, a case study including data of gilthead seabream reared in a range of water temperatures similar to those observed in most of seabream producing countries is used. Herein, data from gilthead seabream reared in Instituto de Acuicultura de Torre de la Sal and presented by Zurita (2013; Table 4.3.1.I. and Fig. 4.3.1.I.) was used to run the model.

Overall the model outputs match well with data presented by Zurita (2013) and can be used for long term forecasting prediction of the final *BW*. As illustrated in Fig. 6-a) the model has an overall trend to overestimate the *BW* data samples. However, there is a deviation of this pattern between days 100 and 200. Removing those 2 data pairs an overall significant ($p < 0.05$; $n = 12$) strong agreement ($r = 0.997$) between the long-term model results and the data is obtained. For the entire growth period (488 days) the estimated mean absolute error (MAE) is 8.8 g.fish^{-1} and the mean absolute percentage error (MAPE) is 8.3%.

Fig. 6-b) illustrates the model application in a scenario whereby the sampled data can be used to correct the model estimates, by restarting the model considering the sampled *BW* instead of the simulated *BW*. The model fit to the data in this scenario improves considerably for the entire growth period ($p < 0.01$; $n = 14$; $r = 0.999$).

After confidence on the model is ensured its outputs can be used to provide valuable insights about the ongoing growing process and feed conversion efficiency. For instance, Fig. 7 presents the monitoring of the daily feed conversion ratio (FCR) along the growth period. The FCR was calculated as the ratio of the feed intake rate data (g.fish^{-1}.day^{-1}) by the simulated weight gain (g.fish^{-1}.day^{-1}). There was an increase of FCR between days 250 and 300. Such model outputs can provide valuable information that can be used as an alert for the farm manager while production is ongoing.

3.3. Model application to simulate seabream growth in a commercial farm

The model applicability to a real case study was evaluated using data from three ponds in a commercial seabream farm (batch 1 to 3 in Fig. 8). The water temperature in the three production datasets ranged between 12.6 °C and 24.8 °C. The model initial conditions and data inputs are presented in Table 2.

Overall the model outputs match well with the production data in the 3 batches (Fig. 8). There is a strong agreement between the model results and the fish *BW* data registered along the experiments (correlation coefficient *r* in all batches is close to one, Table 6). Only in batch

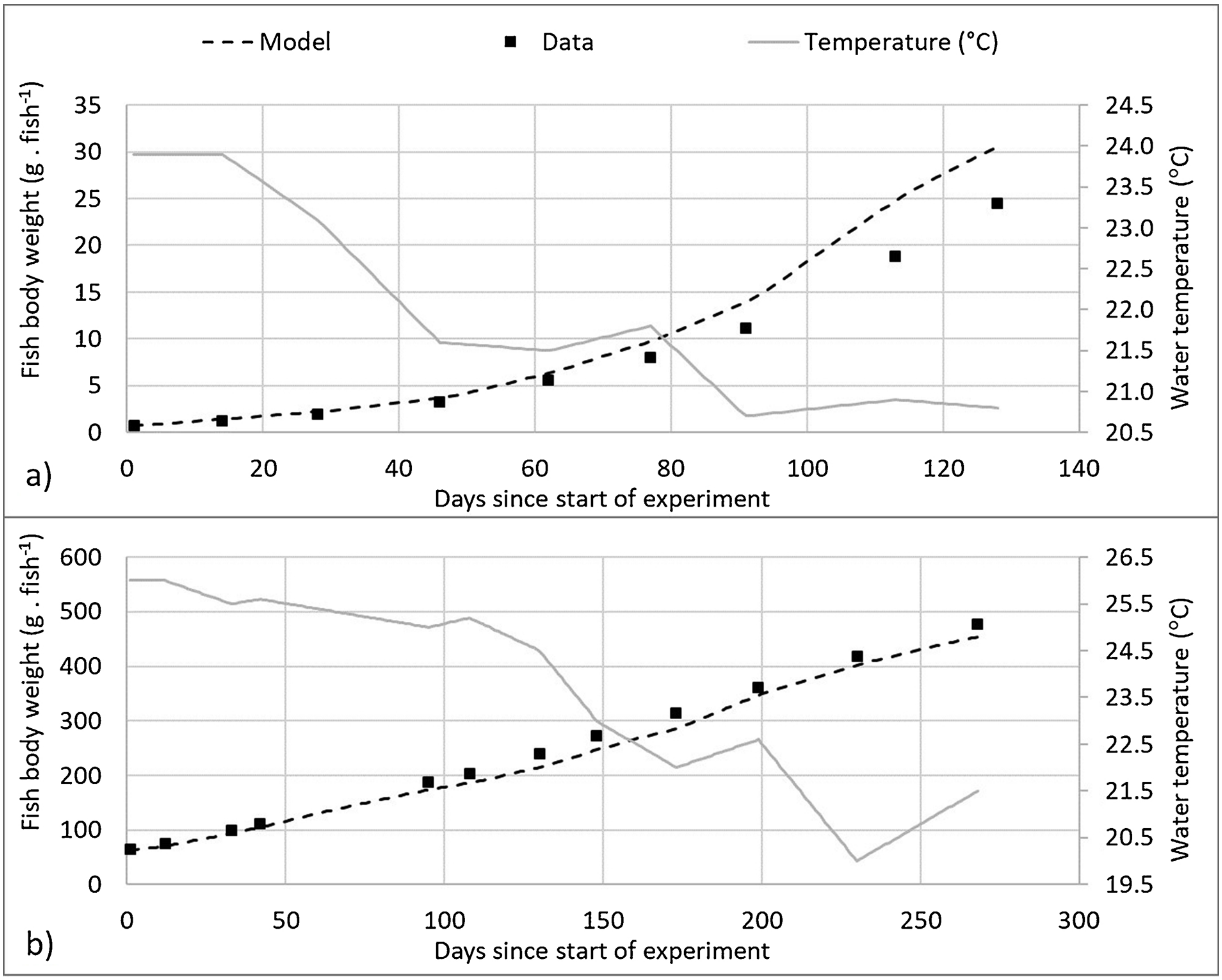

Fig. 3. Fish weight simulation of two datasets from Lupatsch (2003, Annex I): Trial 1 and Trial 12.

3 the statistical significance of the correlation is above the common significance level of 0.05 (p value). In batch 3 the probability of the null hypothesis (i.e. there is no correlation between the data and the model outputs) is around 8%. The maximum absolute error is 21.1 g per fish (*MAE*) and in percentage 8.3% (*MAPE*).

4. Discussion and Conclusions

This paper presents a model for fish weight and body composition simulation that requires a minimal dataset and that can be parameterized based on a set of consistent experiments, the EP model (Energy and protein flux model).

The EP model includes the explicit simulation of protein gains besides energy gain. This is an innovative feature regarding bioenergetic models but still retains the simplification of processes as described by the feed and energy balance shown in Fig. 1. The EP model holds an advantage, compared with the bioenergetic, regarding data inputs and outputs: it also estimates fish body composition over time, besides fish weight, based on feed composition, which is available from the feed labels provided by the manufacturer. The explicit simulation of the fish

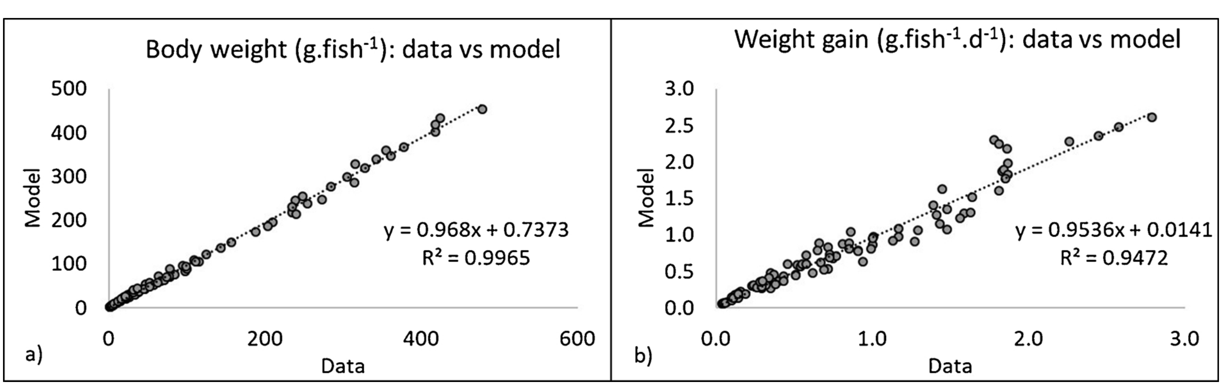

Fig. 4. Model outputs plotted against data: a) body weight (g.fish^{-1}); b) weight gain (g.fish^{-1}.d^{-1}).

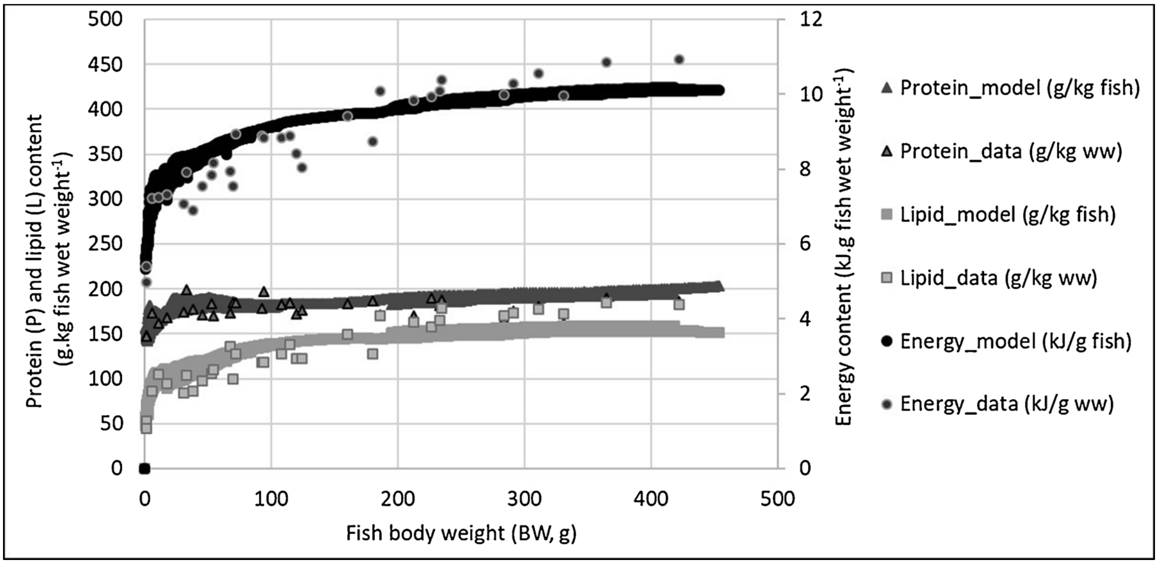

Fig. 5. Protein, lipid and energy content (simulated and data).

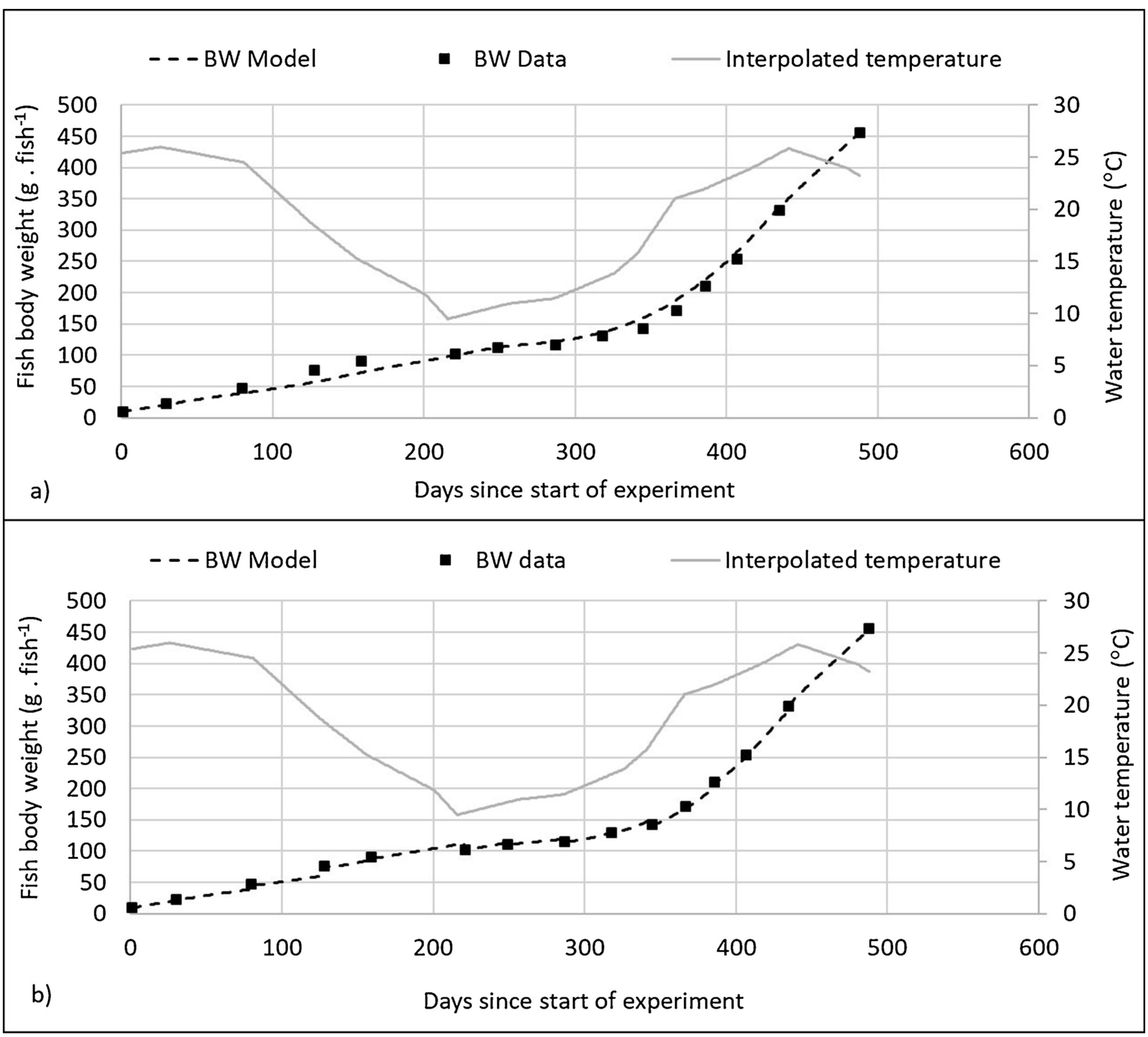

Fig. 6. Fish weight data (Zurita, 2013) and simulation outputs: a) fish growth long term forecasting; b) model correction with sampled data points.

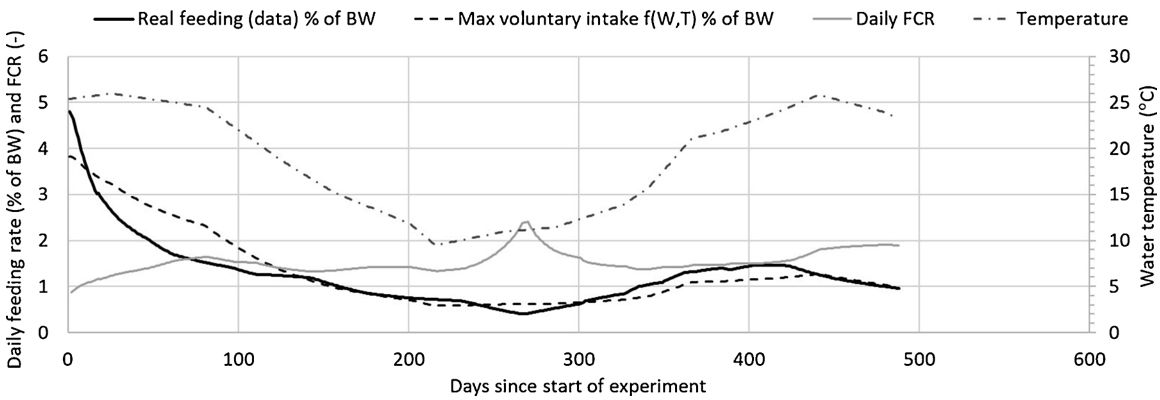

Fig. 7. Feeding outputs of the simulation applied to the Zurita (2013) data: real feeding interpolated from monthly feeding data (here expressed as % of BW), maximum voluntary intake based on Lupatsch (2003) equation for feed intake as a function of temperature and *BW*, daily feed conversion ratio (*FCR*) calculated from the feed data and simulated weight gain.

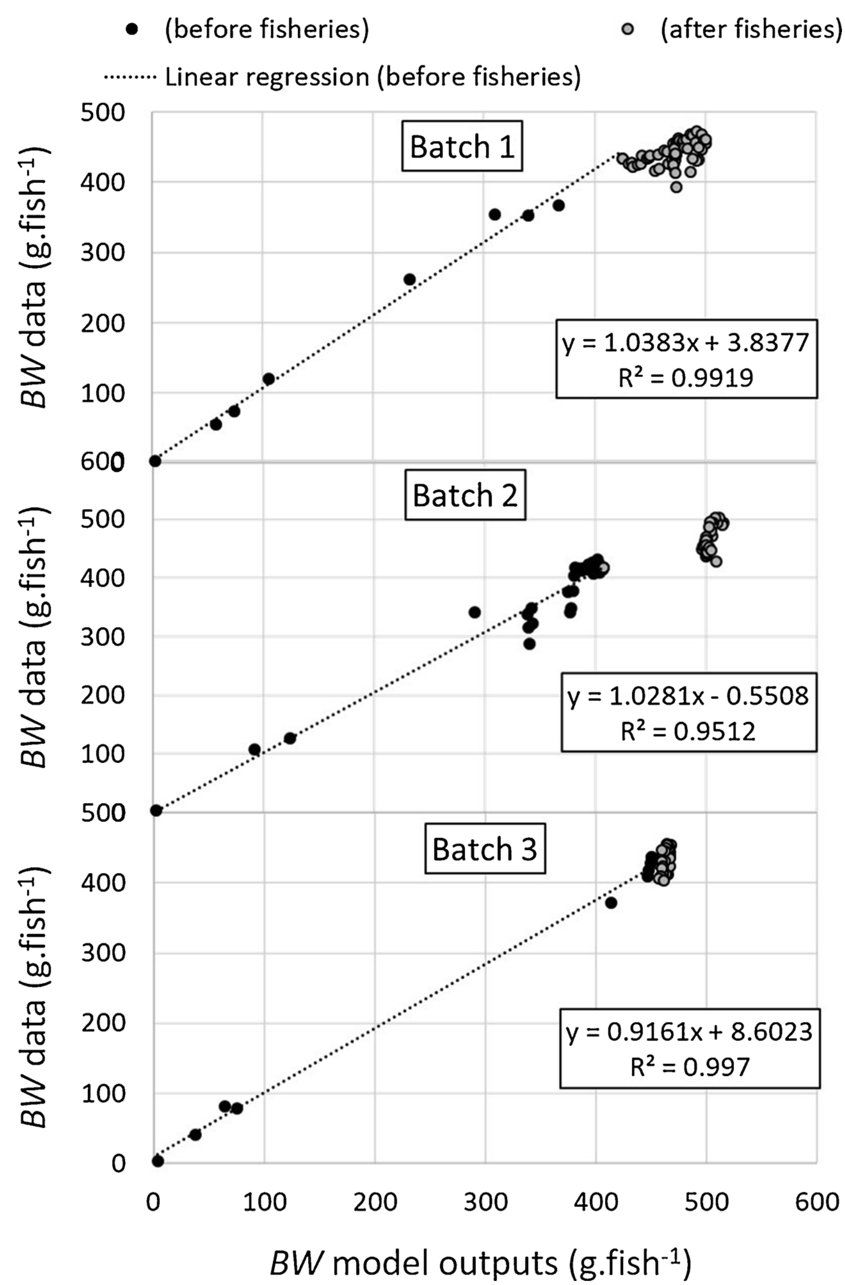

Fig. 8. Model evaluation with fish weight data from a commercial farm.

Table 6
Farm simulation outputs including model evaluation (Pearson's coefficient - *r*, probability of the null hypothesis - *p*, pairs of data-simulated *BW* - *n*, estimated mean absolute error – *MAE*, mean absolute percentage error - *MAPE*).

Farm data:				Model evaluation (before harvesting):						
Batch no.	Initial date	Initial fish no. (x1000)	Initial BW (g)	Final BW (g): data	(simulated)	*r*	*p* (probabil. of null hypothesis)	*n*	*MAE* (g)	*MAPE* (%)
1	11/7/2013	350	2.8	433.4	(425.0)	0.995	0.0497	8	14.2	5.9
2	30/3/2013	310	3.4	417.5	(407.6)	0.957	0.0434	28	21.1	6.0
3	17/8/2012	310	3.7	436.5	(450.3)	0.998	0.0797	8	20.9	8.3

energy and nutrient contents is also an important model feature given that it avoids including it in the model as a decoupled function of *BW*, which is sometimes the origin of unappropriated mathematical formulations as detailed by Canale and Breck (2013).

The simplification of processes by the traditional bioenergetic model (and also encompassed in this EP model) poses several limitations as pointed out elsewhere (Dumas et al., 2010; Jobling, 2011; Nisbet et al., 2012; Stavrakidis-Zachou et al., 2018); however, process simplification is at the same time an asset to implement a fish growth model to simulate on-going production in a commercial farm where field data is limited when compared to data available from scientific experimental settings. The modelling approach presented herein can be implemented in an aquaculture farm with a manageable set of data to force the model (initial *BW*, temperature, feeding rate and feed composition and digestibility). It can be further used for production optimization as for instance illustrated by Seginer and Halachmi (2008) and Seginer and Ben-Asher (2011).

Another significant aspect of the EP model is that its formulation includes only the processes that can be quantified by a set of consistent experiments that lead to the closure of the bioenergetic and protein balance. The parameterization of the EP model based on standard experiments such as the methodology by Lupatsch (2003), Lupatsch et al. (2003a, 2003b), allows to implement a robust model for growth and *BW* composition that is species (or strain) specific. This approach avoids the use of dispersed datasets and substantially reduces the parameter tuning, resulting in a model parameterization that makes sense from a biological standpoint. This work highlights the need for the definition of standards, based on the work by Lupatsch (2003) and Lupatsch et al. (2003a, 2003b), for setting up experiments to parameterize the EP model for other species (or strains).

More research is needed to improve the modelling approach presented herein and to further develop the gilthead seabream model, namely:

i) Carry out an extensive sensitivity analysis of the gilthead seabream EP model.
ii) Include a component for the activity costs, for instance by correlating the water current speed with activity costs for application in gilthead seabream open cages.
iii) Model formulation revision to include maintenance metabolism instead of fasting metabolism. This is a practical requirement for the model to be calibrated, based on experiments that are broadly accepted, as far as fish welfare principles are respected.
iv) *BW* simulation considering 2 compartments following the DEB approach which splits biomass into structure and reserves. This increment in process description will allow for instance that if digestible intake is zero the simulated fish does not immediately lose weight as will happen in this model.
v) Application and validation of the approach to other farmed species.

This approach has limitations, namely whenever more complex process description is required, it might be more suitable to apply a detailed mechanistic approach such as the nutrient-based models. The latter allows simulation of oxygen consumption and ammonia excretion, with a resolution lower than 1 day, and may also assess the effect of different feeding frequencies. Moreover, given the simplification of biological processes (illustrated in Fig. 1) this energy and protein flux model does not simulate the influence of the quality of the digestible nutrients (amino acid profile, fatty acid and lipid class composition), which may be addressed with detailed nutrient-based models. The coupling of the bioenergetic models with nutrient-based models should be further explored to accommodate farms or aquaculture sectors that have available detailed datasets of the ongoing processes, such as the salmon industry. For instance, the following authors developed detailed mechanistic models of different complexity, to estimate the effect of dietary macro-nutrients and amino acid composition on fish growth and body composition: Conceição et al. (1998b) for larval African catfish (*Clarias gariepinus*) and turbot (*Scophthalmus maximus*), Bar and Radde (2009) for Atlantic salmon (*Salmo salar*), Hua et al. (2010) for rainbow trout (*Oncorhynchus mykiss*) and Rønnestad and Conceição (2012) for *Solea senegalensis* larvae.

Acknowledgments

Financial support was provided by the Portuguese Foundation for Science and Technology (FCT) as Postdoc scholarship to Ana Nobre (SFRH/BPD/109442/2015) and by the ValorMar project (24517, 10/SI/2016 - Mobilizing R&TD Programs) co-funded by Compete 2020, Lisboa 2020, Algarve 2020, Portugal 2020 and European Union, through the European Regional Development Fund (ERDF). The authors are grateful to the three anonymous reviewers for valuable comments, which considerably improved this manuscript.

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